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Paper 1: FDA's insights on implementing new strategies for QTc assessment: a review of QT

Bibliographic Information

- **Title:** FDA's insights on implementing new strategies for QTc assessment: a review of QT study reports from 2016 to 2024
- **Authors:** Ji Y, Johannesen L, Garnett C
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Paper Type: GENERIC

Quick Take

This FDA review of 424 QT study reports (2016–2024) documents a 34% decline in thorough QT (TQT) studies and a parallel rise in concentration-QTc (C-QTc) modeling as the primary analysis, driven by ICH E14 Q&A 5.1. Adoption of the white paper C-QTc model and integrated nonclinical risk assessments has substantially reduced sample sizes and streamlined cardiac safety evaluation without compromising regulatory standards.

Executive Summary

The paper summarizes the FDA's experience with evolving ICH E14 strategies for assessing drug-induced QTc prolongation. Using data from 424 QT study reports reviewed between 2016 and 2024, the authors show that the proportion of TQT studies fell from 75% to 41%, while the 5.1 approach (C-QTc modeling in early-phase trials) rose from 6% to 39%. C-QTc as the primary analysis reduced median sample sizes by 35–67% depending on study design. The prespecified linear white paper model was used in 60% of drugs that prolonged QTc. Since 2020, integrated nonclinical risk assessments (double-negative hERG and in vivo QT data) have increasingly supported alternative approaches, especially when clinical exposure multiples are insufficient. The review highlights both successes and challenges, including cases where nonlinear or time-dependent QTc effects required mechanistic understanding beyond simple models. Overall, the modernization of QTc assessment has made trials more efficient while maintaining patient safety.

Scientific Context & Motivation

The ICH E14 guideline (2005) mandated thorough QT studies to exclude a 10 ms QTc prolongation risk. Subsequent Q&A documents (R3 in 2015, final E14/S7B Q&As in 2022) introduced alternative approaches: C-QTc modeling as a primary analysis (Q&A 5.1) and integrated nonclinical risk assessments to supplement clinical data. However, real-world adoption patterns, impact on trial efficiency, and practical challenges had not been systematically quantified from a regulatory perspective. This paper fills that gap by analyzing FDA's internal review database.

✂ Methodological Snapshot

The authors extracted drug, study design, E14 approach, and regulatory outcome data from all QT study reports reviewed by the FDA between January 2016 and August 2024. Descriptive statistics and trends were analyzed using R. For the integrated nonclinical risk assessment experience, reviews from 2020–2024 were examined, focusing on the presence of limitations in clinical, in vitro, or in vivo components. The analysis did not include nonclinical data prior to 2020 because the integrated assessment concept was introduced later.

📄 Detailed Analysis

The paper’s strength lies in its empirical documentation of a major regulatory shift. The 34% reduction in TQT studies is impressive, but the fact that 72% of TQT studies conducted after 2016 did not even request a 5.1 approach suggests that industry adoption could be further accelerated. The high rate of FDA disagreement with 5.1 requests (36/38) highlights persistent challenges, primarily insufficient exposure coverage. The case studies on macimorelin and oliceridine are particularly valuable: they illustrate that a simple linear C-QTc model can fail when the mechanism is not hERG-mediated or when time-dependent effects exist. This underscores the need for dose-ranging data and mechanistic understanding before relying on model-based extrapolation. The integrated nonclinical risk assessment framework is a major step forward, but the finding that 50% of clinical data and 33% of in vitro assays had limitations indicates that high-quality execution remains critical. The paper also implicitly raises the question of whether the 10 ms threshold remains the optimal regulatory benchmark in an era of richer data.

Domain Context

This work sits at the intersection of clinical pharmacology, regulatory science, and cardiac safety. It reflects the maturation of model-informed drug development, where physiologically based pharmacokinetics, concentration-response modeling, and nonclinical assays are combined to replace dedicated clinical trials. The ICH E14/S7B Q&As represent a rare example of guidelines evolving in response to scientific advances, and this paper provides the evidentiary basis for that evolution from the FDA’s perspective.

🔑 Key Findings

1. Between 2016 and 2023, TQT studies decreased from 75% to 41% of QT reports, while 5.1 approach studies increased from 6% to 39%.
2. C-QTc as primary analysis reduced median sample sizes by 67% (parallel), 42% (nested crossover), and 35% (crossover) compared to by-time analysis.
3. Among 70 QTc-prolonging drugs, 60% used the white paper linear C-QTc model; nonlinear models (log-linear, E_{max}) were used in 16%.

4. Integrated nonclinical risk assessments were used in 51 reviews (2020–2024), mostly supporting the 5.1 approach; limitations were identified in 50% of clinical data and 33% of in vitro assays.
 5. Case studies (nirmatrelvir, macimorelin, oliceridine, gepirone) illustrate successful use of double-negative nonclinical data and pitfalls of simple C-QTc models when mechanisms are complex.
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Strengths & Limitations

Strengths

- Large, comprehensive dataset of 424 QT study reports spanning 8 years of FDA reviews.
- Provides a unique regulatory perspective on the real-world impact of ICH E14 Q&A changes.
- Quantifies efficiency gains (sample size reduction) and adoption trends with clear visualizations.
- Includes instructive case studies that highlight both successes and remaining challenges.

Limitations (Acknowledged by Authors)

- Data are limited to FDA submissions; findings may not fully reflect global practice.
- The analysis did not systematically track the frequency with which 5.1 requests were rejected by FDA.
- Nonclinical data before 2020 were excluded due to different methodology, so the full historical picture is incomplete.
- The study did not evaluate how often nonclinical signals translated into clinical QTc prolongation.

Limitations (Expert Review)

- Retrospective design and reliance on sponsor-submitted reports may introduce selection bias.
- No code or data are publicly available, limiting independent verification.
- Generalizability to non-US regulatory environments is uncertain.
- The paper does not assess the impact on overall drug development timelines or costs beyond sample size.

Generalizability

The trends and efficiency gains are likely applicable to other ICH regions that have adopted the same Q&A documents, though local regulatory practices may differ. The case studies provide broadly relevant lessons for C-QTc modeling.

Figures & Tables

- **Figure 1:** Flowchart summarizing the 424 QT study reports by E14 approach (TQT, 5.1, 6.1) and outcome.
 - *Significance:* Provides an overview of the dataset composition and the relative frequency of each regulatory pathway.

- **Figure 2:** Trends in the percentage of TQT studies and 5.1 approach studies by year of FDA review (2016–2023).
 - *Significance:* Visually demonstrates the 34% decline in TQT studies and the corresponding rise of the 5.1 approach.
 - **Figure 4:** Cumulative distribution of sample sizes for TQT studies using by-time vs. C-QTc primary analysis, stratified by study design.
 - *Significance:* Quantifies the substantial sample size savings achieved by adopting C-QTc as the primary analysis.
 - **Table 2:** Comparison of key features (controls, exposure coverage, criteria for negative study) across TQT, 5.1, and 6.1 approaches.
 - *Significance:* A concise reference for understanding the regulatory expectations and differences among the three pathways.
 - **Table 3:** Case study of nirmatrelvir (PAXLOVID) illustrating successful use of a double-negative integrated nonclinical risk assessment to support the 5.1 approach.
 - *Significance:* Shows a real-world example where nonclinical data compensated for insufficient clinical exposure multiples.
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Future Directions

Further refinement of C-QTc models for drugs with hysteresis or time-dependent effects; integration of in silico proarrhythmia models (e.g., CiPA) into regulatory decision-making; prospective studies to validate the predictive value of double-negative nonclinical assessments; and extension of these approaches to large-molecule and targeted therapies.

Expert Commentary

This paper is a landmark summary from the FDA’s QT Interdisciplinary Review Team, documenting a successful regulatory science evolution. The data-driven demonstration that C-QTc modeling and nonclinical integration can safely replace many dedicated TQT studies should encourage wider adoption. The cautionary tales (macimorelin, oliceridine) remind us that modeling must be guided by pharmacology, not just statistics. As we move toward model-informed drug development, this experience provides a template for how guidelines can adapt without compromising patient safety.

Bottom Line

The shift from resource-intensive TQT studies to C-QTc modeling and integrated nonclinical risk assessments has made cardiac safety evaluation more efficient without sacrificing rigor. Sponsors should embrace the 5.1 approach when feasible, use the white paper linear model as a default, and be prepared to justify more complex models with mechanistic data. The double-negative nonclinical framework is now a viable pathway to supplement clinical exposure gaps.

☒ Figures

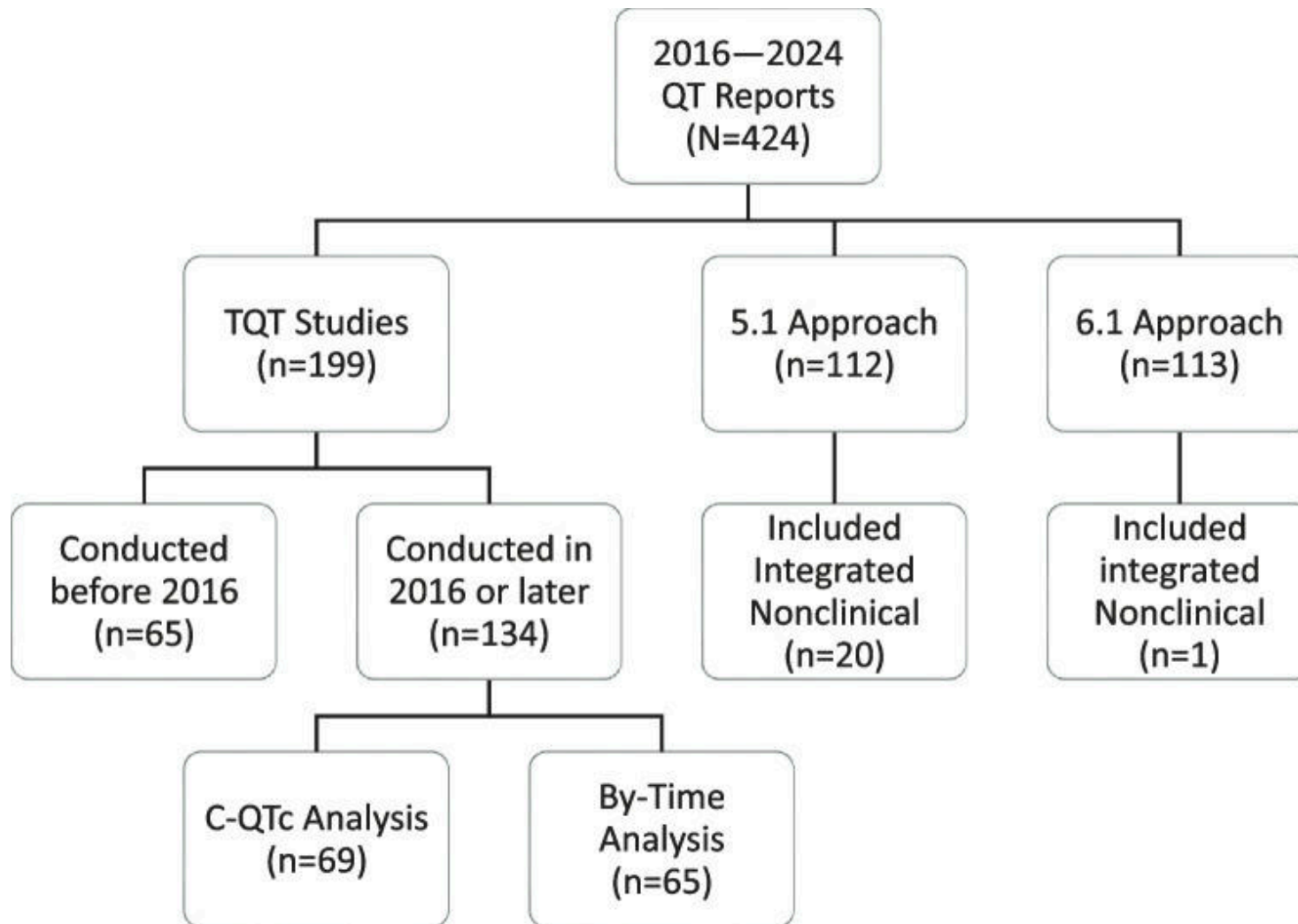


Figure 1: Summary of 424 QT study reports reviewed by FDA between 2016 and 2024

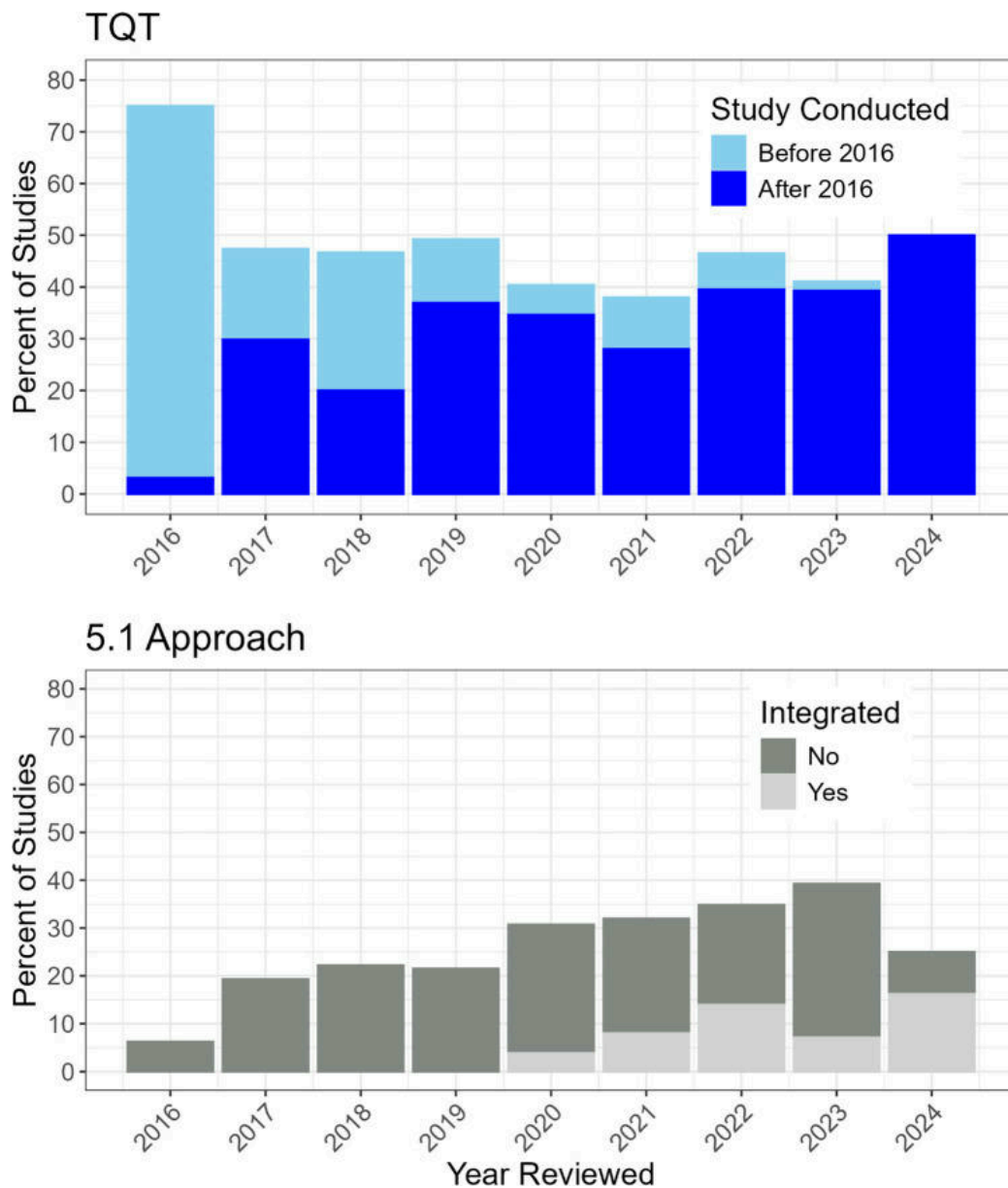


Figure 2: Percentage of TQT studies (Top) and studies following the 5.1 approach (bottom) by the year reviewed by FDA

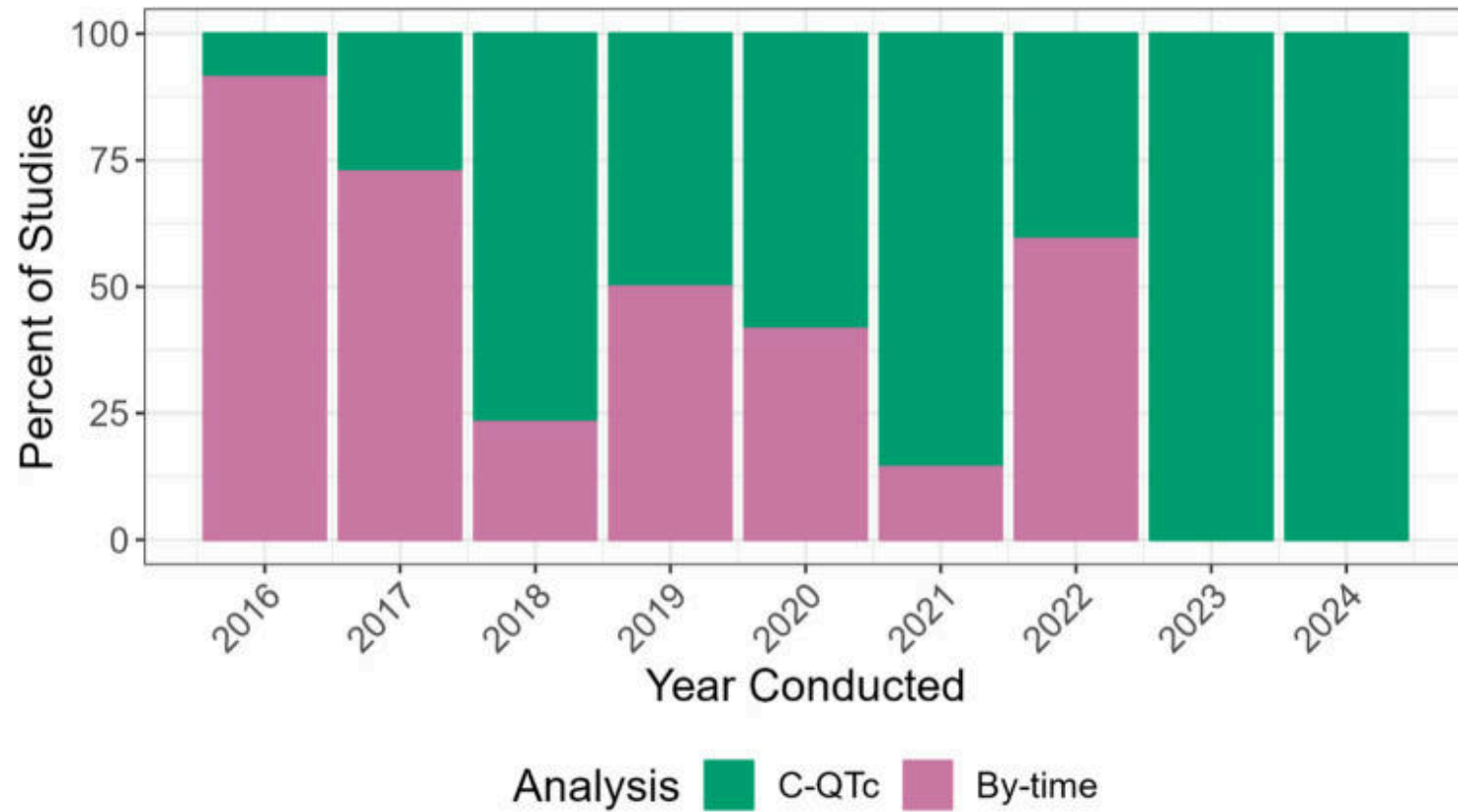


Figure 3: Percentage of analysis methods of TQT studies by the year study conducted

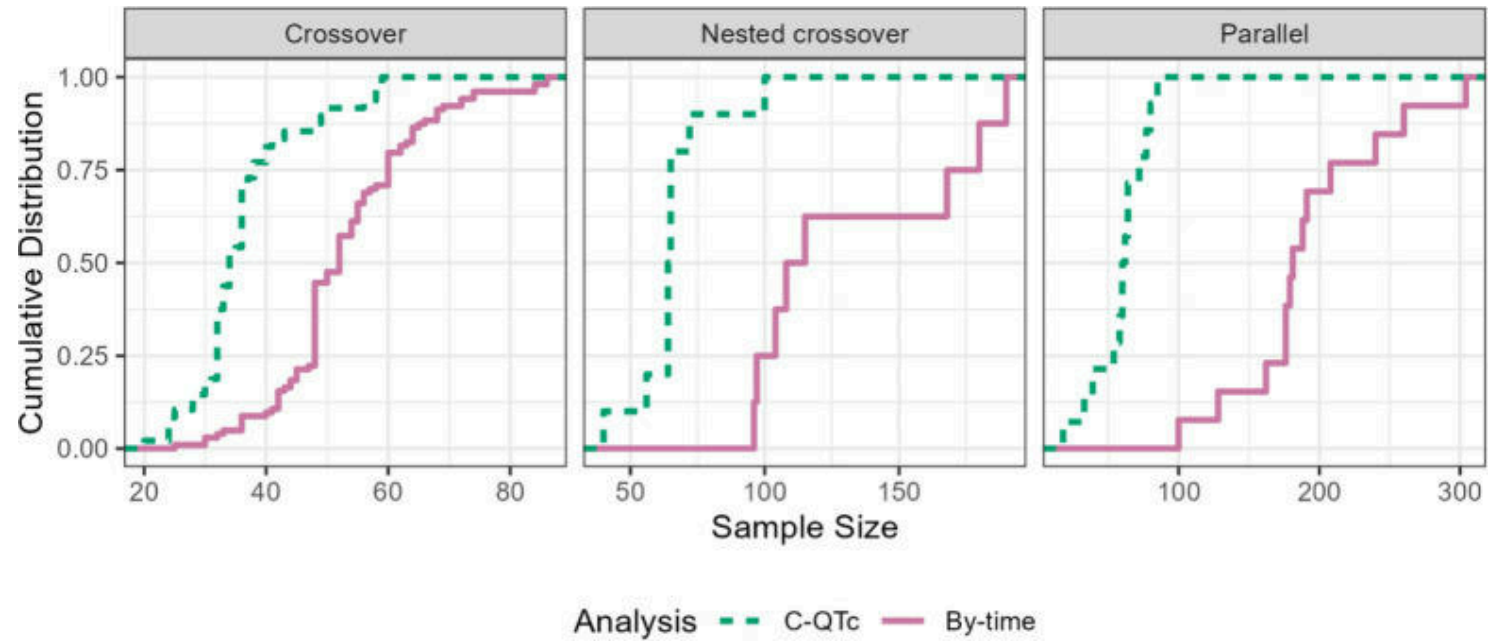


Figure 4: Cumulative distribution of sample size by analysis methods and study designs